

Fourier Multiplication Circuit

Discovered in a 1-Layer Transformer Trained on $(a + b) \bmod 113$

Auto-generated by Grokking Circuit Discovery Tool

June 1, 2026

Circuit Summary

Task: $(a + b) \bmod 113$

FVE: 0.9384

Formula Accuracy: 100.00%

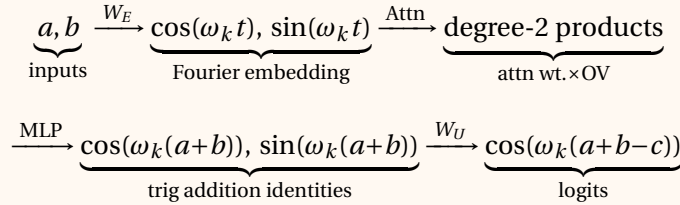
Key frequencies: $\mathcal{K} = \{1, 20, 25\}$ (3 frequencies)

Neurons assigned: 0/512 ($k = 1: 0, k = 20: 0, k = 25: 0$)

The Core Equation

$$\text{Logit}(c \mid a, b) = \sum_{k \in \mathcal{K}} \underbrace{\alpha_k}_{\text{learned amplitude}} \cdot \underbrace{\cos\left(\frac{2\pi k(a+b-c)}{113}\right)}_{\substack{\text{max at } c=(a+b) \bmod 113 \\ \text{(constructive interference)}}} \quad (1)$$

Data Flow Pipeline



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1 Discovered Circuit: Fourier Multiplication Algorithm

The model computes $(a + b) \bmod 113$ using 3 key frequencies $k \in \{1, 20, 25\}$ with angular frequencies $\omega_k = \frac{2\pi k}{113}$.

1.1 Step 1: Embedding (Token $\rightarrow$ Fourier Components)

The embedding matrix W_E maps each one-hot token $t \in \{0, \dots, 112\}$ onto sinusoidal components:

$$\mathbf{x}_a^{(0)} = \underbrace{W_E \cdot \mathbf{e}_a}_{\text{token embedding of } a} + \underbrace{\mathbf{p}_0}_{\text{positional embedding (pos 0)}} \quad (2)$$

$$\approx \sum_{k \in \mathcal{K}} \left[\underbrace{\alpha_k \cos\left(\frac{2\pi k \cdot a}{113}\right)}_{\text{cosine component from } W_E \text{ row } a} \cdot \mathbf{u}_k^{(\cos)} + \underbrace{\beta_k \sin\left(\frac{2\pi k \cdot a}{113}\right)}_{\text{sine component from } W_E \text{ row } a} \cdot \mathbf{u}_k^{(\sin)} \right] \quad (3)$$

where $\mathcal{K} = \{1, 20, 25\}$ are the key frequencies discovered via Fourier analysis of W_E .

Similarly for token b at position 1:

$$\mathbf{x}_b^{(0)} = \underbrace{W_E \cdot \mathbf{e}_b}_{\text{token embedding of } b} + \underbrace{\mathbf{p}_1}_{\text{positional embedding (pos 1)}} \quad (4)$$

1.2 Step 2: Attention (Move Fourier Info to Output Position)

The attention heads compute scores from the '=' token (position 2) to positions 0 and 1, then produce a weighted combination:

$$\text{score}_{\Rightarrow a}^{(j)} = \frac{\overbrace{\mathbf{x}_{=}^{(0)} \cdot W_Q^{(j)}}^{\text{query from '=' position}} \cdot \overbrace{\left(W_K^{(j)}\right)^\top \cdot \mathbf{x}_a^{(0)}}^{\text{key from position } a}}{\underbrace{\sqrt{d_{\text{head}}}}_{\text{scaling factor}}} \quad (5)$$

$$A_0^{(j)} = \underbrace{\sigma\left(\text{score}_{\Rightarrow a}^{(j)} - \text{score}_{\Rightarrow b}^{(j)}\right)}_{\substack{\text{softmax over 2 elements} \\ = \text{sigmoid of difference}}} \quad (6)$$

$$\approx \underbrace{0.5}_{\text{uniform baseline}} + \underbrace{\gamma_j \left(\cos(\omega_{k_j} \cdot a) - \cos(\omega_{k_j} \cdot b) \right)}_{\substack{\text{periodic modulation at frequency } k_j \\ \text{(from } C^{(j)} = W_E^\top W_K^{(j)\top} W_Q^{(j)} \mathbf{x}_{=}^{(0)})}} \quad (7)$$

$$\text{attn_out}^{(j)} = \underbrace{A_0^{(j)}}_{\text{attn weight to } a} \cdot \underbrace{W_O^{(j)} W_V^{(j)} \mathbf{x}_a^{(0)}}_{\text{OV circuit applied to } a} + \underbrace{A_1^{(j)}}_{\text{attn weight to } b} \cdot \underbrace{W_O^{(j)} W_V^{(j)} \mathbf{x}_b^{(0)}}_{\text{OV circuit applied to } b} \quad (8)$$

Key insight: Since $A_0^{(j)}$ is approximately linear in $\cos(\omega_{k_j} a)$ and the OV circuit outputs $\cos(\omega_{k_j} a)$, their product creates *degree-2 polynomials* of sines and cosines — exactly what is needed for the trig identity in the next step.

1.3 Step 3: MLP (Compute Trig Identities)

The residual stream after attention contains degree-2 products. The MLP neurons compute the key trigonometric identities:

$$\mathbf{x}^{(1)} = \underbrace{\mathbf{x}^{(0)}}_{\substack{\text{skip connection} \\ (= \text{token embedding})}} + \sum_{j=0}^3 \underbrace{\text{attn_out}^{(j)}}_{\substack{\text{attention head } j \\ \text{output (degree-2 trig)}}} \quad (9)$$

$$\text{pre}_n = \underbrace{W_{\text{in}}[n, :] \cdot \mathbf{x}^{(1)}}_{\substack{\text{linear projection of neuron } n}} + \underbrace{b_{\text{in}}[n]}_{\text{bias}} \quad (10)$$

$$\text{MLP}[n] = \underbrace{\text{ReLU}(\text{pre}_n)}_{\substack{\text{activation of neuron } n \\ \approx \text{degree-2 polynomial of } \cos(\omega_k a), \sin(\omega_k a), \dots}} \quad (11)$$

$$(12)$$

The core computation: For each key frequency k , the MLP neurons collectively compute:

$$\underbrace{\mathbf{u}_k^\top \cdot \text{MLP}(a, b)}_{\substack{\text{projection of MLP activations} \\ \text{onto direction } \mathbf{u}_k \text{ in } W_L}} \approx \underbrace{\alpha_k \cos(\omega_k(a+b))}_{\text{target: cosine of sum}} \quad (13)$$

$$= \overbrace{\alpha_k \cos(\omega_k a) \cos(\omega_k b) - \alpha_k \sin(\omega_k a) \sin(\omega_k b)}^{\text{cosine addition formula}} \quad (14)$$

from neurons computing $\cos(\omega_k a) \cos(\omega_k b)$
from neurons computing $\sin(\omega_k a) \sin(\omega_k b)$

$$\underbrace{\mathbf{v}_k^\top \cdot \text{MLP}(a, b)}_{\substack{\text{projection onto} \\ \text{direction } \mathbf{v}_k \text{ in } W_L}} \approx \underbrace{\beta_k \sin(\omega_k(a+b))}_{\text{target: sine of sum}} \quad (15)$$

$$= \overbrace{\beta_k \sin(\omega_k a) \cos(\omega_k b) + \beta_k \cos(\omega_k a) \sin(\omega_k b)}^{\text{sine addition formula}} \quad (16)$$

from neurons computing $\sin(\omega_k a) \cos(\omega_k b)$
from neurons computing $\cos(\omega_k a) \sin(\omega_k b)$

where $\mathbf{u}_k, \mathbf{v}_k \in \mathbb{R}^{512}$ are the directions in the neuron-logit map $W_L = W_{\text{out}}^\top W_U^\top$ corresponding to $\cos(\omega_k c)$ and $\sin(\omega_k c)$ respectively.

1.4 Step 4: Unembedding (Fourier $\rightarrow$ Logits)

The neuron-logit map $W_L = W_{\text{out}}^\top \cdot W_U^\top$ converts the MLP activations to output logits:

$$W_L \approx \sum_{k \in \mathcal{K}} \left[\underbrace{\cos(\omega_k c)}_{\substack{\text{vector in } \mathbb{R}^P \\ (\text{c-th entry} = \cos(\omega_k c))}} \cdot \underbrace{\mathbf{u}_k^\top}_{\substack{\text{direction in neuron space} \\ \text{that reads } \cos(\omega_k(a+b))}} + \underbrace{\sin(\omega_k c)}_{\substack{\text{vector in } \mathbb{R}^P \\ (\text{c-th entry} = \sin(\omega_k c))}} \cdot \underbrace{\mathbf{v}_k^\top}_{\substack{\text{direction in neuron space} \\ \text{that reads } \sin(\omega_k(a+b))}} \right] \quad (17)$$

$$(18)$$

Therefore, the logit for output class c is:

$$\text{Logit}(c \mid a, b) = \underbrace{W_L \cdot \text{MLP}(a, b)}_{\substack{\text{neuron-logit map applied to MLP output}}} \quad (19)$$

$$\approx \sum_{k \in \mathcal{K}} \left[\underbrace{\cos(\omega_k c)}_{\substack{\text{from } W_L \\ \text{(unembed direction)}}} \cdot \underbrace{\mathbf{u}_k^\top \text{MLP}(a, b)}_{\substack{\text{MLP computes this} \\ \approx \alpha_k \cos(\omega_k (a+b))}} + \underbrace{\sin(\omega_k c)}_{\substack{\text{from } W_L \\ \text{(unembed direction)}}} \cdot \underbrace{\mathbf{v}_k^\top \text{MLP}(a, b)}_{\substack{\text{MLP computes this} \\ \approx \beta_k \sin(\omega_k (a+b))}} \right] \quad (20)$$

$$\approx \sum_{k \in \mathcal{K}} \alpha_k \underbrace{\left[\cos(\omega_k c) \cos(\omega_k (a+b)) + \sin(\omega_k c) \sin(\omega_k (a+b)) \right]}_{\text{cosine subtraction identity}} \quad (21)$$

$$= \sum_{k \in \mathcal{K}} \underbrace{\alpha_k \cos(\omega_k (a+b-c))}_{\substack{\text{peaks when } c \equiv a+b \pmod{113} \\ \text{since } \cos(0)=1 \text{ is maximum}}} \quad (22)$$

1.5 Step 5: Prediction via Constructive Interference

$$\hat{c} = \underbrace{\arg \max_{c \in \{0, \dots, 112\}}}_{\text{select class with highest logit}} \overbrace{\sum_{k \in \mathcal{K}} \alpha_k \cos\left(\frac{2\pi k(a+b-c)}{113}\right)}^{\text{sum of cosines at key frequencies}} \quad (23)$$

$$= \underbrace{(a+b) \bmod 113}_{\substack{\text{constructive interference:} \\ \text{all cosines equal 1 when } c \equiv a+b \pmod{113} \\ \text{destructive interference elsewhere}}} \quad (24)$$

1.6 Why Multiple Frequencies? (Constructive Interference)

A single cosine $\cos(\omega_k x)$ has period 113 but near-maxima at other values of x . By summing over multiple key frequencies, the model creates a function with a *unique* maximum at $x = 0 \bmod 113$:

$$f(x) = \sum_{k \in \mathcal{K}} \underbrace{\cos\left(\frac{2\pi k \cdot x}{113}\right)}_{\text{each has max at } x=0} \quad (25)$$

$$f(0) = \underbrace{3}_{\text{all 3 cosines}=1} \gg f(x \neq 0) \quad \underbrace{(\text{destructive interference})}_{\text{cosines at different frequencies cancel}} \quad (26)$$

1.7 Worked Example: $a = 7, b = 13$

Step 1: Embed. Compute Fourier components for each key frequency k :

$$\begin{aligned} k = 1: & \cos\left(\frac{2\pi \cdot 1 \cdot 7}{113}\right), \sin\left(\frac{2\pi \cdot 1 \cdot 7}{113}\right), \cos\left(\frac{2\pi \cdot 1 \cdot 13}{113}\right), \sin\left(\frac{2\pi \cdot 1 \cdot 13}{113}\right) \\ k = 20: & \cos\left(\frac{2\pi \cdot 20 \cdot 7}{113}\right), \sin\left(\frac{2\pi \cdot 20 \cdot 7}{113}\right), \cos\left(\frac{2\pi \cdot 20 \cdot 13}{113}\right), \sin\left(\frac{2\pi \cdot 20 \cdot 13}{113}\right) \\ k = 25: & \cos\left(\frac{2\pi \cdot 25 \cdot 7}{113}\right), \sin\left(\frac{2\pi \cdot 25 \cdot 7}{113}\right), \cos\left(\frac{2\pi \cdot 25 \cdot 13}{113}\right), \sin\left(\frac{2\pi \cdot 25 \cdot 13}{113}\right) \end{aligned}$$

Step 2: MLP applies trig identity $\cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta) = \cos(\alpha + \beta)$:

$$\begin{aligned} k = 1: & \underbrace{\cos\left(\frac{2\pi \cdot 1 \cdot 7}{113}\right) \cos\left(\frac{2\pi \cdot 1 \cdot 13}{113}\right) - \sin\left(\frac{2\pi \cdot 1 \cdot 7}{113}\right) \sin\left(\frac{2\pi \cdot 1 \cdot 13}{113}\right)}_{=\cos\left(\frac{2\pi \cdot 1 \cdot 20}{113}\right)} \\ k = 20: & \underbrace{\cos\left(\frac{2\pi \cdot 20 \cdot 7}{113}\right) \cos\left(\frac{2\pi \cdot 20 \cdot 13}{113}\right) - \sin\left(\frac{2\pi \cdot 20 \cdot 7}{113}\right) \sin\left(\frac{2\pi \cdot 20 \cdot 13}{113}\right)}_{=\cos\left(\frac{2\pi \cdot 20 \cdot 20}{113}\right)} \end{aligned}$$

Step 3: Logit for correct answer $c^* = (7 + 13) \bmod 113 = 20$:

$$\text{Logit}(20) = \sum_{k \in \mathcal{K}} \alpha_k \underbrace{\cos\left(\omega_k \cdot \overbrace{(7 + 13 - 20)}^{=0 \bmod 113}\right)}_{=\cos(0)=1} = \sum_k \alpha_k \quad (\text{MAXIMUM})$$

For any $c \neq 20$, the cosines at different frequencies point in different directions and **destructively interfere**, giving a smaller logit.

1.7.1 Frequency $k = 1$: $\omega_1 = \frac{2\pi \cdot 1}{113}$

Embedding:

$$\underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{DFT}} \underbrace{\cos\left(\frac{2\pi \cdot 1 \cdot a}{113}\right)}_{\text{component } k=1}, \underbrace{\sin\left(\frac{2\pi \cdot 1 \cdot a}{113}\right)}_{\text{component } k=1}$$

Attention:

$$\begin{aligned} A_0^{(j)} &\approx \underbrace{0.5 + \gamma_j \cos(\omega_1 a)}_{\text{head } j \text{ tuned to freq 1}} \times \underbrace{\text{OV}^{(j)}(a)}_{\text{OV output at freq 1}} \approx \cos(\omega_1 a) \\ &\Rightarrow \underbrace{\cos^2(\omega_1 a), \cos(\omega_1 a) \cos(\omega_1 b), \dots}_{\text{degree-2 products from attn} \times \text{OV}} \end{aligned}$$

MLP (trig identity):

$$\begin{aligned} &\underbrace{\text{neurons } n \in \text{cluster}_1}_{0 \text{ neurons at freq 1}} \xrightarrow{\text{ReLU}} \\ &\quad \underbrace{\cos(\omega_1 a) \cos(\omega_1 b) - \sin(\omega_1 a) \sin(\omega_1 b)}_{\substack{\text{from attn} \quad \text{from attn}}} \\ &\quad \quad \quad = \cos(\omega_1 (a+b)) \end{aligned}$$

Unembedding:

$$\begin{aligned} &\underbrace{\mathbf{u}_1^\top \cdot \text{MLP}}_{\text{project in } W_L} \approx \alpha_1 \cos(\omega_1 (a+b)) \\ &\quad \xrightarrow{\times \cos(\omega_1 c)} \underbrace{\alpha_1 \cos(\omega_1 (a+b-c))}_{\text{freq 1 contribution to logit}(c)} \end{aligned}$$

1.7.2 Frequency $k = 20$: $\omega_{20} = \frac{2\pi \cdot 20}{113}$

Embedding:

$$\underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{DFT}} \underbrace{\cos\left(\frac{2\pi \cdot 20 \cdot a}{113}\right)}_{\text{component } k=20}, \underbrace{\sin\left(\frac{2\pi \cdot 20 \cdot a}{113}\right)}_{\text{component } k=20}$$

Attention:

$$\begin{aligned} A_0^{(j)} &\approx \underbrace{0.5 + \gamma_j \cos(\omega_{20} a)}_{\text{head } j \text{ tuned to freq 20}} \times \underbrace{\text{OV}^{(j)}(a)}_{\text{OV output at freq 20}} \approx \cos(\omega_{20} a) \\ &\Rightarrow \underbrace{\cos^2(\omega_{20} a), \cos(\omega_{20} a) \cos(\omega_{20} b), \dots}_{\text{degree-2 products from attn} \times \text{OV}} \end{aligned}$$

MLP (trig identity):

$$\begin{aligned}
 & \underbrace{\text{neurons } n \in \text{cluster}_{20}}_{0 \text{ neurons at freq } 20} \xrightarrow{\text{ReLU}} \\
 & \quad = \cos(\omega_{20}(a+b)) \\
 & \quad \underbrace{\cos(\omega_{20}a) \cos(\omega_{20}b)}_{\text{from attn}} - \underbrace{\sin(\omega_{20}a) \sin(\omega_{20}b)}_{\text{from attn}}
 \end{aligned}$$

Unembedding:

$$\begin{aligned}
 & \underbrace{\mathbf{u}_{20}^\top \cdot \text{MLP}}_{\text{project in } W_L} \approx \alpha_{20} \cos(\omega_{20}(a+b)) \\
 & \quad \xrightarrow{\times \cos(\omega_{20}c)} \underbrace{\alpha_{20} \cos(\omega_{20}(a+b-c))}_{\text{freq 20 contribution to logit}(c)}
 \end{aligned}$$

1.7.3 Frequency $k = 25$: $\omega_{25} = \frac{2\pi \cdot 25}{113}$

Embedding:

$$\underbrace{W_E[a]}_{\text{row } a \text{ of } W_E} \xrightarrow{\text{DFT}} \underbrace{\cos\left(\frac{2\pi \cdot 25 \cdot a}{113}\right)}_{\text{component } k=25}, \underbrace{\sin\left(\frac{2\pi \cdot 25 \cdot a}{113}\right)}_{\text{component } k=25}$$

Attention:

$$\begin{aligned}
 & \underbrace{A_0^{(j)} \approx 0.5 + \gamma_j \cos(\omega_{25}a)}_{\text{head } j \text{ tuned to freq } 25} \times \underbrace{\text{OV}^{(j)}(a) \approx \cos(\omega_{25}a)}_{\text{OV output at freq } 25} \\
 & \Rightarrow \underbrace{\cos^2(\omega_{25}a), \cos(\omega_{25}a) \cos(\omega_{25}b), \dots}_{\text{degree-2 products from attn} \times \text{OV}}
 \end{aligned}$$

MLP (trig identity):

$$\begin{aligned}
 & \underbrace{\text{neurons } n \in \text{cluster}_{25}}_{0 \text{ neurons at freq } 25} \xrightarrow{\text{ReLU}} \\
 & \quad = \cos(\omega_{25}(a+b)) \\
 & \quad \underbrace{\cos(\omega_{25}a) \cos(\omega_{25}b)}_{\text{from attn}} - \underbrace{\sin(\omega_{25}a) \sin(\omega_{25}b)}_{\text{from attn}}
 \end{aligned}$$

Unembedding:

$$\begin{aligned}
 & \underbrace{\mathbf{u}_{25}^\top \cdot \text{MLP}}_{\text{project in } W_L} \approx \alpha_{25} \cos(\omega_{25}(a+b)) \\
 & \quad \xrightarrow{\times \cos(\omega_{25}c)} \underbrace{\alpha_{25} \cos(\omega_{25}(a+b-c))}_{\text{freq 25 contribution to logit}(c)}
 \end{aligned}$$